

A penalized IMEX finite volume scheme for the isentropic Euler system at all Mach numbers

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We present a finite volume scheme for the solution of the isentropic Euler system that works uniformly at all Mach numbers. The method employs a flux-splitting approach to handle implicitly the stiff nonlinear pressure term, which is linearized using a penalization technique, while retaining an explicit discretization for the convective contribution. The scheme is asymptotic-preserving in the low Mach number limit, and can also deal with shock waves at high Mach numbers. Numerical results demonstrate the accuracy and robustness of the scheme.

Keywords: Isentropic Euler equations; IMplicit-EXplicit scheme; Penalization method; Asymptotic Preserving; Low Mach.

1. Introduction

The compressible and incompressible flow regimes exhibit significantly different characteristics, requiring different numerical methods to accurately capture their behavior. Several numerical schemes have been developed to independently handle each regime, with applications to various physical phenomena. However, in certain scenarios encountered in environmental, nuclear or aerospace applications, the Mach number may vary both spatially and temporally. Directly coupling incompressible and compressible solvers is highly challenging, making the development of unified schemes, which are capable of dealing with flows across all Mach numbers, an active area of research.

A major challenge in designing all-Mach numerical schemes is avoiding the restrictive time step imposed by the CFL stability condition, which depends on the sound speed. This requires an efficient discretization of stiff terms. A common strategy is the IMplicit-EXplicit (IMEX) approach¹, which treats the stiff terms implicitly, while keeping the remaining terms explicit. Moreover, this technique automatically guarantees a correct scal-

ing of the numerical viscosity in the low Mach regime. In the case of the isentropic Euler equations, the stiffness arises from the pressure term, which is a nonlinear function on the density, making direct computation expensive. Various approaches have been explored in the literature to address this issue²⁻⁸. In Ref. 6 a nonlinear equation for the pressure is solved, whereas in Ref. 7,8 a semi-implicit linearization is forwarded, based on the first-order derivative of the pressure term, and on an *ad hoc* parameter which combines low and high Mach flux contributions.

In this work, we propose a novel linearization approach, based on the penalization strategy originally introduced in Ref. 9 for kinetic equations. A flux splitting technique¹⁰ is then carried out, which avoids the use of problem-dependent parameters, and is suitable for an IMEX time marching technique. The resulting numerical scheme is proven to be asymptotic-preserving in the low Mach limit.

The remainder of the paper is organized as follows. In Section 2, the isentropic Euler system is introduced, along with its nondimensional formulation. Section 3 details the proposed numerical scheme, and its asymptotic preserving property. Numerical results are provided in Section 4, followed by conclusion remarks in Section 5.

2. Mathematical model

Let $\Omega \in \mathbb{R}$ be a bounded one-dimensional domain, defined by the spatial position $x \in \mathbb{R}$ and the temporal variable $t \in \mathbb{R}_+$. We consider the isentropic Euler system of equations:

$$\begin{aligned}\partial_t \rho + \partial_x q &= 0, \\ \partial_t q + \partial_x(qu) + \partial_x p &= 0.\end{aligned}\tag{1}$$

In this formulation, ρ is the density, u is the velocity, $q = \rho u$ is the momentum, and p is the pressure. In particular, the equation of state for the pressure depends on the density as $p(\rho) = C\rho^\gamma$, where $C > 0$ is a positive constant, and $\gamma \geq 1$ is the polytropic index of the gas.

To derive the low Mach number limit of the model, the non-dimensional formulation is necessary. Introducing the reference quantities with the "0" subscript, the following dimensionless variables are defined:

$$\tilde{x} = \frac{x}{L_0}, \quad \tilde{u} = \frac{u}{u_0}, \quad \tilde{t} = \frac{t}{\tau_0} = \frac{|u_0|}{L_0}t, \quad \tilde{\rho} = \frac{\rho}{\rho_0},$$

for which the scaling of the temporal variable holds considering a unit Strouhal number $St = \frac{L_0}{\tau_0|u_0|}$. Defining also the sound speed as $c = \sqrt{\frac{\partial p}{\partial \rho}} =$

$\sqrt{C\gamma\rho^{\gamma-1}}$, we can introduce a stiffness parameter $\varepsilon = \gamma M^2$, based on the reference Mach number, $M = |u_0|/c_0$. Inserting these expressions in (1), and omitting the tilde for readability, the non-dimensional system reads as:

$$\begin{aligned}\partial_t \rho + \partial_x q &= 0, \\ \partial_t q + \partial_x(qu) + \frac{1}{\varepsilon} \partial_x(C\rho^\gamma) &= 0.\end{aligned}\tag{2}$$

The scaled system shows that in the low Mach limit, for $\varepsilon \rightarrow 0$, the pressure term is stiff. The eigenvalues of the system are $\lambda = [u, u \pm c/\sqrt{\varepsilon}]$, so that the time step for a completely explicit scheme would be limited by a CFL-type stability condition, which, for a mesh of characteristic size Δx , is given by

$$\Delta t \leq \text{CFL} \min_{\Omega} \left(\frac{\Delta x}{\max(|\lambda|)} \right) = \mathcal{O}(\sqrt{\varepsilon}),\tag{3}$$

that yields prohibitively small time steps for $\varepsilon \rightarrow 0$.

3. Numerical scheme

The proposed numerical scheme uses the flux splitting technique introduced by Toro and Vázquez¹⁰ for the full Euler system. To explain it, we can rewrite system (2) in the general form as:

$$\partial_t \mathbf{Q} + \partial_x \mathbf{F} = \mathbf{0},\tag{4}$$

where $\mathbf{Q} = (\rho, q)^T$ is the vector of conserved variables, and the fluxes are $\mathbf{F} = (q, qu + C\rho^\gamma/\varepsilon)^T$. The nonlinear flux vector can be split into a stiff and a non-stiff part, that will be discretized implicitly and explicitly, respectively:

$$\begin{aligned}\mathbf{F} &= \mathbf{F}_{\text{ex}} + \mathbf{F}_{\text{im}}, \\ &= \begin{bmatrix} 0 \\ qu \end{bmatrix} + \begin{bmatrix} q \\ \frac{C}{\varepsilon} \rho^\gamma \end{bmatrix}.\end{aligned}\tag{5}$$

In this way, the time step does not depend anymore on the Mach regime, but only on the material speed, thus we have

$$\Delta t \leq \text{CFL} \min_{\Omega} \left(\frac{\Delta x}{|\lambda^c|} \right),\tag{6}$$

with $\lambda^c = 2u$ being the eigenvalues of the convective sub-system

$$\partial_t \mathbf{Q} + \partial_x \mathbf{F}_{\text{ex}} = \mathbf{0}.$$

In the following, we first present the semi-discrete scheme in time, and provide a proof of its Asymptotic Preserving (AP) property in the low Mach number limit, and then discuss the fully discrete scheme.

3.1. First order semi-discrete scheme in time

The non-dimensional system (2) is discretized in time by the following implicit-explicit scheme:

$$\rho^{n+1} = \rho^n - \Delta t \partial_x q^{n+1}, \quad (7a)$$

$$q^{n+1} = q^n - \Delta t \partial_x (qu)^n - \frac{\Delta t}{\varepsilon} \partial_x (C\rho^\gamma)^{n+1}. \quad (7b)$$

Although the stiffness arises only from the pressure term in the momentum equation, the advection term in the continuity equation is also treated implicitly. This choice is necessary to ensure the asymptotic preserving property, hence retrieving a wave equation for the density, and to prevent the scheme to be constrained by the stability condition (3). The above system is solved by substitution, hence inserting the momentum equation (7b) into the continuity equation (7a) leads to a wave equation for the density:

$$\rho^{n+1} = \rho^n - \Delta t \partial_x q^n + \Delta t^2 \partial_{xx} (qu)^n + \frac{\Delta t^2}{\varepsilon} \partial_x (\partial_x (C\rho^\gamma)^{n+1}). \quad (8)$$

Due to the nonlinear pressure term, the above system for the unknown density is nonlinear, which may require a high computational effort to be solved. To circumvent this problem, we propose to use a penalization technique⁹ that allows us to rewrite the pressure flux as

$$\partial_x (C\rho^\gamma)^{n+1} = \partial_x (C\rho^\gamma)^n - \partial_x \rho^n + \partial_x \rho^{n+1}, \quad (9)$$

so now the momentum equation can be rewritten as:

$$q^{n+1} = q^n - \Delta t \partial_x (qu)^n - \frac{\Delta t}{\varepsilon} (\partial_x (C\rho^\gamma - \rho)^n + \partial_x \rho^{n+1}). \quad (10)$$

The resulting semi-discrete scheme is then given by (7a)-(10), and the wave equation (8) now becomes:

$$\rho^{n+1} - \frac{\Delta t^2}{\varepsilon} \partial_{xx} \rho^{n+1} = \rho^n - \Delta t \partial_x q^n + \Delta t^2 \partial_{xx} (qu)^n + \frac{\Delta t^2}{\varepsilon} \partial_x (\partial_x (C\rho^\gamma - \rho)^n). \quad (11)$$

This equation is now linear and is first solved for ρ^{n+1} and then used in (10) to solve for q^{n+1} , at time $t^{n+1} = t^n + \Delta t$.

3.2. Asymptotic preserving property

Here, we want to prove the AP property of the numerical scheme, in the limit $\varepsilon \rightarrow 0$. The proof is firstly conducted at the continuous level, then at the semi-discrete level. Let us assume a generic bounded domain Ω , with

boundary $\partial\Omega$, and $\mathbf{n}$ the unit outward normal vector to such boundary. We can define impermeable boundary conditions as $\mathbf{u} \cdot \mathbf{n} = 0$. If a generic variable is called m , which here represents either the density or the momentum, the k -th order power expansion, in terms of the parameter ε , is written as $m = m_{(0)} + \varepsilon m_{(1)} + \varepsilon^2 m_{(2)} + \dots + \mathcal{O}(\varepsilon^k)$. Assuming well-prepared initial data, the expansion can be inserted in the system of equations (2) for each variable. From the momentum equation, one immediately obtains

$$\mathcal{O}(\varepsilon^{-1}) : \quad \partial_x \rho_{(0)}^\gamma = 0 \rightarrow \rho_{(0)} = \rho_{(0)}(t). \quad (12)$$

At the 0-th leading order, the entire system reads as:

$$\begin{aligned} \mathcal{O}(\varepsilon^0) : \quad & \partial_t \rho_{(0)} + \partial_x (q_{(0)}) = 0, \\ & \partial_t q_{(0)} + \partial_x (q_{(0)} u_{(0)}) + C \partial_x \rho_{(1)}^\gamma = 0. \end{aligned} \quad (13)$$

Starting from the continuity equation in (13), taking its integral over the entire domain, and applying Gauss theorem for the second term, we obtain:

$$\int_{\Omega} \partial_t \rho_{(0)} dx + \int_{\partial\Omega} q_{(0)} \cdot \mathbf{n} dS = \int_{\Omega} \partial_t \rho_{(0)} dx + \int_{\partial\Omega} \rho_{(0)} u_{(0)} \cdot \mathbf{n} dS = 0, \quad (14)$$

which simplifies, using the previous result (12) and the impermeable boundary conditions, to $\Omega \partial_t \rho_0 = 0$. Therefore, the density is a constant, both in space and time. Such outcome can be exploited in the continuity equation again, finally providing the incompressibility constraint $\rho_{(0)} \partial_x u_{(0)} = 0$ at the continuous level.

Passing to the semi-discrete level, the system (7a)-(10) may be separated into an explicit and an implicit sub-system, following the flux splitting strategy. The explicit sub-system writes

$$\rho^* = \rho^n, \quad (15a)$$

$$q^* = q^n - \Delta t \partial_x (qu)^n - \frac{\Delta t}{\varepsilon} \partial_x (C \rho^\gamma - \rho)^n, \quad (15b)$$

and, consequently, the implicit sub-system is given by

$$\rho^{n+1} = \rho^* - \Delta t \partial_x q^{n+1}, \quad (16a)$$

$$q^{n+1} = q^* - \frac{\Delta t}{\varepsilon} \partial_x \rho^{n+1}. \quad (16b)$$

Substituting the ε -power expansion in the explicit sub-system, the order $\mathcal{O}(\varepsilon^{-1})$ yields

$$\partial_x (C \rho_{(0)}^\gamma - \rho_{(0)})^n = 0, \quad (17)$$

which implies that $\rho_{(0)}^n$ and $(\rho_{(0)}^\gamma)^n$ correspond, unless a constant. The same order $\mathcal{O}(\varepsilon^{-1})$ for the implicit sub-system leads to

$$\partial_x \rho_{(0)}^{n+1} = 0, \quad (18)$$

meaning that the new density at the 0-th leading order is constant in space. Taking now the integral form of the continuity equation (16a) at order $\mathcal{O}(\varepsilon^0)$, we obtain:

$$\int_{\Omega} \rho_{(0)}^{n+1} dx = \int_{\Omega} \rho_{(0)}^n dx - \Delta t \int_{\partial\Omega} \rho_{(0)}^{n+1} u_{(0)}^{n+1} \cdot \mathbf{n} dS, \quad (19)$$

and using the impermeability boundary conditions we have that $\rho_{(0)}^{n+1} = \rho_{(0)}^n$. Taking again the same continuity equation and using this last result in addition to (18), we are left with $\rho_{(0)}^{n+1} \partial_x u_{(0)}^{n+1} = 0$, that is a consistent discretization of the incompressibility constraint previously retrieved at the continuous level.

3.3. First order fully discrete scheme in time and space

For the discretization in space of the domain $\Omega = [x_{\max}, x_{\min}]$, of length $L = x_{\max} - x_{\min}$, a regular grid of N elements is chosen, with dimension $\Delta x = L/N$. Each element can be identified by its center x_i , and faces $x_{i\pm 1/2}$. A finite volume scheme is used to solve the explicit sub-system:

$$\rho_i^* = \rho_i^n, \quad (20a)$$

$$q_i^* = q_i^n - \frac{\Delta t}{\Delta x} \left(\hat{f}_{i+1/2}^{ex} - \hat{f}_{i-1/2}^{ex} \right), \quad (20b)$$

where $\hat{f}_{i\pm 1/2}$ is a Rusanov flux function¹¹ for the explicit fluxes $f^{ex} = qu + (C\rho^\gamma - \rho)/\varepsilon$. The numerical dissipation is only proportional to $|2u|$, hence vanishing for $\varepsilon \rightarrow 0$. The implicit sub-system, which yields the wave equation (11), is discretized in space by central finite difference operators:

$$\rho_i^{n+1} - \frac{\Delta t^2}{\varepsilon \Delta x^2} (\rho_{i+1}^{n+1} - 2\rho_i^{n+1} + \rho_{i-1}^{n+1}) = \rho_i^n - \frac{\Delta t}{2\Delta x} (q_{i+1}^* - q_{i-1}^*). \quad (21)$$

Eq. (21) is a tridiagonal linear system that can be solved for ρ_i^{n+1} with the Thomas algorithm. Next, the momentum is updated according to (16b):

$$q_i^{n+1} = q_i^* - \frac{\Delta t}{2\varepsilon \Delta x} (\rho_{i+1}^{n+1} - \rho_{i-1}^{n+1}). \quad (22)$$

Remark 3.1 (Additional numerical dissipation). *The implicit sub-system does not embed any numerical dissipation. However, for high Mach*

number flows, an additional artificial viscosity term might be needed to prevent spurious oscillations to arise. This can be efficiently taken into account by adding an implicit Laplace operator in the linear system (21), that is

$$\rho_i^{n+1} - (1 + \nu) \frac{\Delta t^2}{\varepsilon \Delta x^2} (\rho_{i+1}^{n+1} - 2\rho_i^{n+1} + \rho_{i-1}^{n+1}) = \rho_i^n - \frac{\Delta t}{2\Delta x} (q_{i+1}^* - q_{i-1}^*), \quad (23)$$

where $\nu = \max_{\Omega} |2u| \Delta x$ is the artificial dissipation coefficient.

Remark 3.2 (Second order extensions in space and time). *As the space discretization is done using central finite difference and finite volume schemes on a regular Cartesian grid, second order of accuracy is automatically achieved, provided that the numerical dissipation in the flux evaluation of (20b) is computed with a total variation diminishing (TVD) reconstruction technique. For the time discretization, first order is presented in this work, but second order may be easily reached by means of a multi-step IMEX Runge-Kutta scheme¹.*

4. Numerical results

In this section, some numerical results are presented, covering a wide range of Mach numbers, to show the versatility of the proposed penalization technique. The novel scheme is in the following referred as P-IMEX (Penalized-Implicit Explicit). Unless specified differently, the isentropic constant is $C = 1$ and the polytropic index is $\gamma = 2$ for all the test cases. Finally, the CFL number is $\text{CFL} = 0.9$.

4.1. Convergence study

For the spatial convergence study, a manufactured solution method is applied, which consists in defining a source term $\mathbf{S}$, that balances the fluxes of the original system (5):

$$\mathbf{Q}_t + \mathbf{F}_x = \mathbf{S}, \quad \mathbf{S} = \begin{bmatrix} \partial_x q \\ \partial_x (qu + \frac{p}{\varepsilon}) \end{bmatrix}. \quad (24)$$

In this case, the computational domain is $\Omega = [-2.5, 2.5]$, discretized with a sequence of successively refined meshes of N elements, and characterized by periodic boundary conditions. The initial condition writes

$$\rho(x, 0) = \left(1 + \frac{(\gamma - 1)u}{2\sqrt{\gamma}}\right)^2, \quad u(x, 0) = \sin\left(\frac{2\pi x}{L}\right), \quad (25)$$

and the stiffness parameter is varied as $\varepsilon = \{1, 10^{-2}, 10^{-4}\}$. Errors of the conserved variables are measured in L_2 norm at the final time $t_f = 0.1$, and are presented in Table 1 for increasing mesh refinement, showing the expected second order of accuracy in space. Results are compared to the linearization strategy forwarded in Ref. 8 for the same test case, exhibiting slightly smaller errors for the new P-IMEX method.

Table 1. Numerical convergence results for the manufactured solution with different stiffness regimes using the novel penalized IMEX finite volume scheme. The errors for density and momentum are given in L_2 norm at the final time $t_f = 0.1$.

N	$L_2(\rho)$	$\mathcal{O}(\rho)$	$L_2(q)$	$\mathcal{O}(q)$
$\varepsilon = 1$				
25	6.9999E-03	-	3.6993E-03	-
50	1.6855E-03	2.05	9.2637E-04	2.00
100	4.1388E-04	2.03	2.4961E-04	1.89
200	1.0264E-04	2.01	6.5482E-05	1.93
$\varepsilon = 10^{-2}$				
25	9.1014E-04	-	1.1362E-02	-
50	2.3384E-04	1.96	3.8636E-03	1.56
100	6.3161E-05	1.89	1.1069E-03	1.80
200	1.6718E-05	1.92	2.9477E-04	1.91
$\varepsilon = 10^{-4}$				
25	9.3017E-06	-	1.8098E-02	-
50	1.0589E-06	3.13	3.6382E-03	2.31
100	2.0740E-07	2.35	1.0158E-03	1.84
200	2.3509E-08	3.14	2.5888E-04	1.97

4.2. Riemann problem from Degond-Tang

In Ref. 8, a test case consisting of several Riemann problems is proposed, with initial condition:

$$\rho(x, 0) = \begin{cases} 1 & \text{if } x \in [0, 0.2] \\ 1 + \varepsilon & \text{if } x \in (0.2, 0.3] \\ 1 & \text{if } x \in (0.3, 0.7] \\ 1 - \varepsilon & \text{if } x \in (0.7, 0.8) \\ 1 & \text{if } x \in [0.8, 1] \end{cases}, \quad q(x, 0) = \begin{cases} 1 - \frac{\varepsilon}{2} & \text{if } x \in [0, 0.2] \\ 1 & \text{if } x \in (0.2, 0.3] \\ 1 + \frac{\varepsilon}{2} & \text{if } x \in (0.3, 0.7] \\ 1 & \text{if } x \in (0.7, 0.8) \\ 1 - \frac{\varepsilon}{2} & \text{if } x \in [0.8, 1] \end{cases}. \quad (26)$$

Periodic boundary conditions are considered over the domain $\Omega = [0, 1]$, and a constant spatial discretization using $N = 200$ elements is cho-

Table 2. Numerical convergence results for the manufactured solution with different stiffness regimes using the semi-implicit finite volume scheme of Ref. 8. The errors for density and momentum are given in L_2 norm at the final time $t = 0.1$.

N	$L_2(\rho)$	$\mathcal{O}(\rho)$	$L_2(q)$	$\mathcal{O}(q)$
$\varepsilon = 1$				
25	6.4408E-03	-	6.3690E-03	-
50	1.6004E-03	2.01	1.4587E-03	2.13
100	3.9950E-04	2.00	3.5957E-04	2.02
200	9.9930E-05	2.00	9.0068E-05	2.00
$\varepsilon = 10^{-2}$				
25	5.8285E-04	-	1.3733E-02	-
50	1.7971E-04	1.70	4.0898E-03	1.75
100	5.4247E-05	1.73	1.1417E-03	1.84
200	1.5452E-05	1.81	3.0085E-04	1.92
$\varepsilon = 10^{-4}$				
25	1.1881E-05	-	1.7189E-02	-
50	1.5080E-06	2.98	4.1965E-03	2.03
100	1.8982E-07	2.99	1.0397E-03	2.01
200	2.6615E-08	2.83	2.6040E-04	2.00

sen. The solution is compared to a reference solution obtained with $N = 1000$ at the final time $t_f = 0.05$, for three different stiffness values $\varepsilon = \{0.8^2, 0.3^2, 0.05^2\}$, corresponding to the compressible, intermediate and incompressible regimes. The results are depicted in Fig. 1, demonstrating an overall good agreement with the reference and the literature⁷.

4.3. Two colliding acoustic waves

In this test case⁸, we simulate two colliding acoustic waves in the domain $\Omega = [-1, 1]$, with periodic boundary conditions, and $N = 100$. The flow is initially assigned as

$$\rho(x, 0) = 0.955 + \frac{\varepsilon}{2}(1 - \cos(2\pi x)), \quad u(x, 0) = -\text{sign}(x)\sqrt{\gamma}(1 - \cos(2\pi x)). \quad (27)$$

with $\gamma = 1.4$ and $\varepsilon = 0.1^2$. The numerical solution for density and momentum is plot in Fig. 2 for different times t , and a comparison against the reference solution obtained with $N = 1000$ cells is shown. No spurious oscillations are generated and the wave patterns are correctly captured.

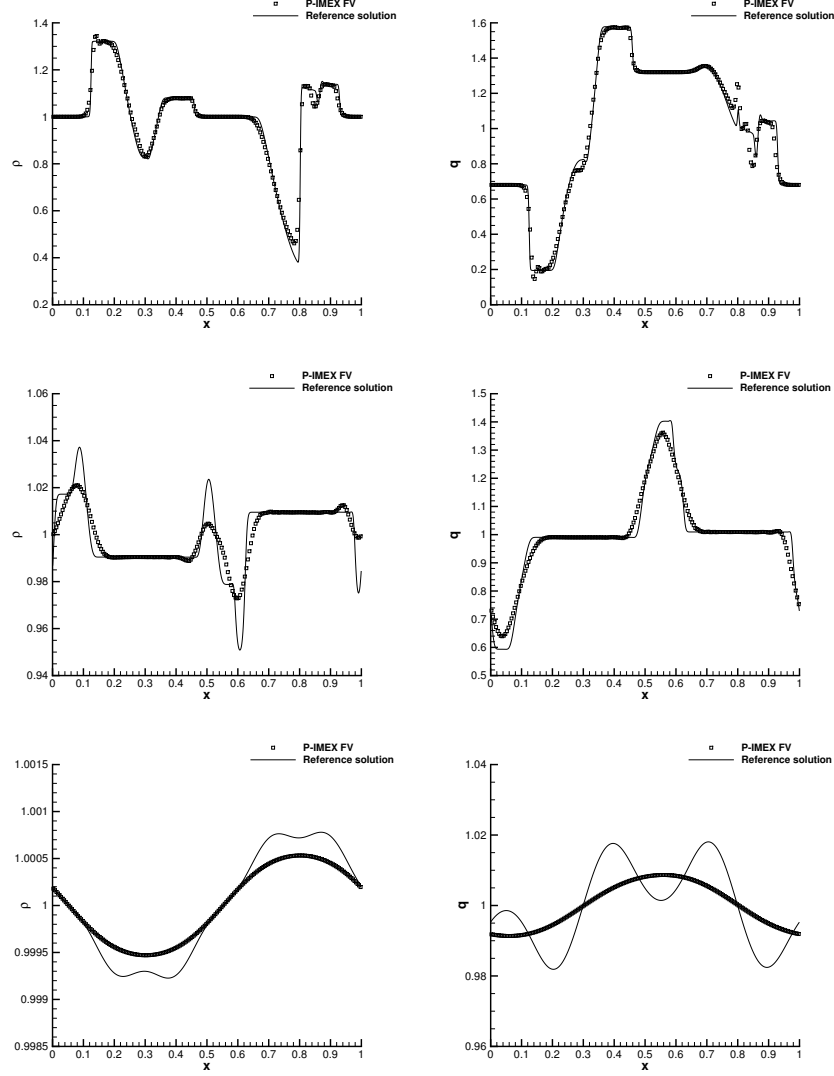


Fig. 1. Riemann problem from Degond-Tang at the final time with $N = 200$ for $\varepsilon = 0.8^2$ (top row), $\varepsilon = 0.3^2$ (middle row) and $\varepsilon = 0.05^2$ (bottom row). Left column: density. Right column: momentum.

4.4. Riemann problem at high Mach number

This Riemann problem² is characterized by the following initial condition:

$$\rho(x, 0) = \begin{cases} 3 & \text{if } x \in [0, 0.5] \\ 1 & \text{if } x \in (0.5, 1] \end{cases}, \quad q(x, 0) = 0, \quad (28)$$

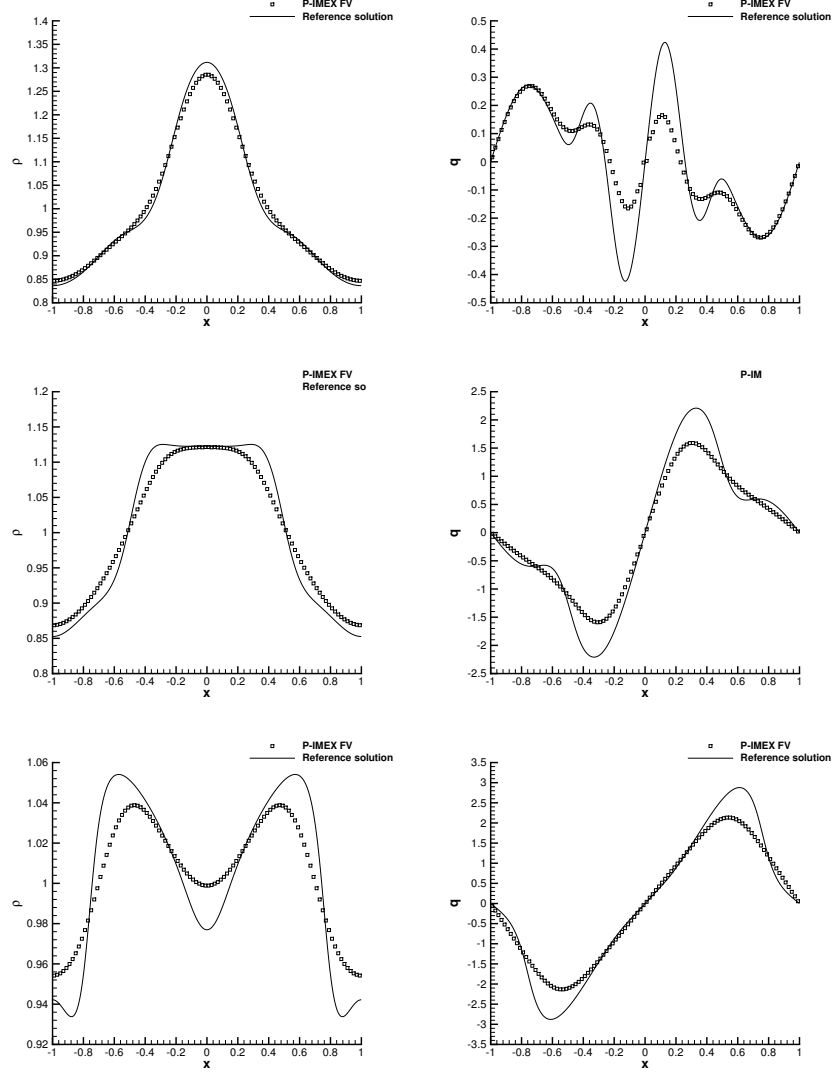


Fig. 2. Two colliding acoustic waves with $N = 100$ cells and $\varepsilon = 0.01$ at output time $t = 0.04$ (top row), $t = 0.06$ (middle row) and $t = 0.08$ (bottom row). Left column: density. Right column: momentum.

prescribed over the domain $\Omega = [0, 1]$, with reflective boundary conditions, $N = 100$, and the equation of state defined with $\gamma = 1.4$. Two sets of stiff-

ness parameter and final time are considered: $[\varepsilon, t_f] = \{[1, 0.16], [400, 1.8]\}$. We are in the compressible regime, as the Mach number is pretty high, but the P-IMEX scheme shows good performances in Fig. 3, where a comparison against the reference solution computed on $N = 1000$ elements is proposed.

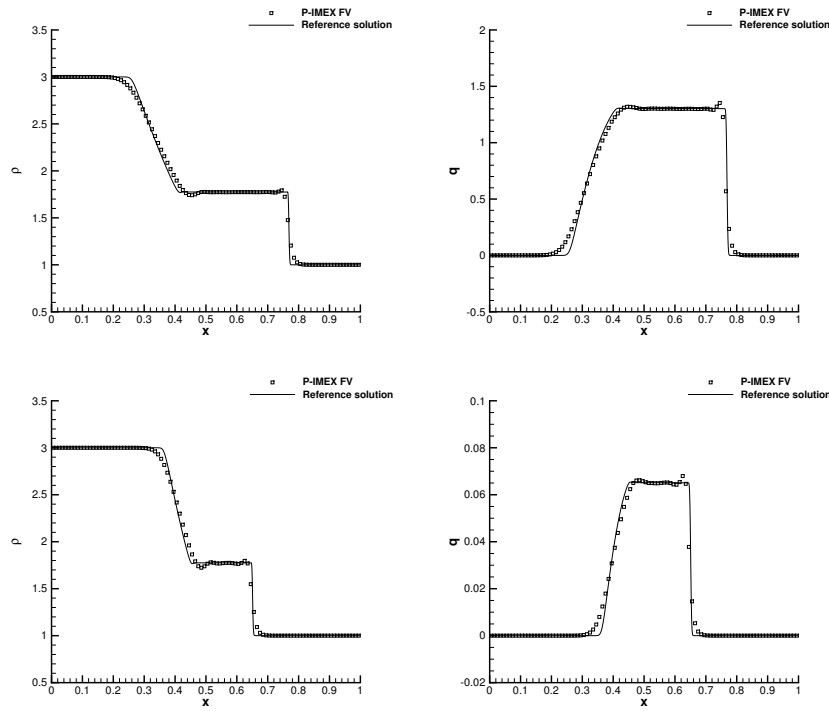


Fig. 3. Riemann problem at high Mach number with $N = 100$. Top: $\varepsilon = 1$ and $t = 0.16$. Bottom: $\varepsilon = 400$ $t = 1.8$. Left column: density. Right column: momentum.

Finally, a strong shock wave problem⁷ is considered:

$$\rho(x, 0) = \begin{cases} 10 \\ 20 \end{cases}, \quad q(x, 0) = \begin{cases} 2000 & \text{if } x \in [0, 0.5] \\ 0 & \text{if } x \in (0.5, 1] \end{cases}, \quad (29)$$

with $\gamma = 1.4$, $N = 800$ and $\varepsilon = 1$. The results, depicted in Fig. 4, are in agreement with the literature⁷.

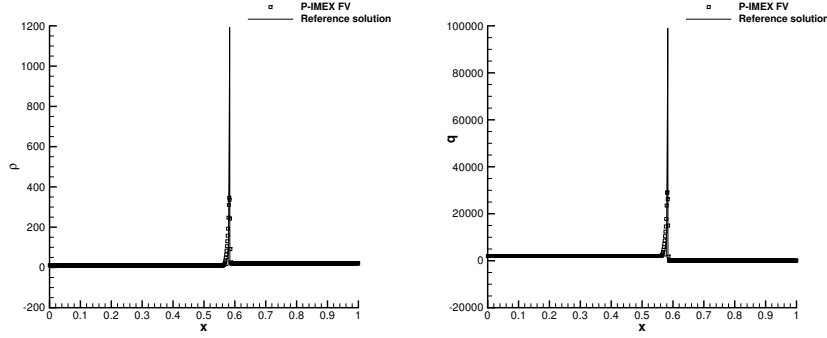


Fig. 4. Strong shock wave problem $N = 800$ at $t_f = 0.001$.

4.5. Double rarefaction test problem

Here, two rarefaction waves are considered⁶ with initial condition:

$$\rho(x, 0) = \begin{cases} 1 + \varepsilon & \text{if } x \in [0, 0.5] \\ 1 & \text{if } x \in (0.5, 1] \end{cases}, \quad q(x, 0) = \begin{cases} (1 + \varepsilon)(1 - \sqrt{\varepsilon}) & \text{if } x \in [0, 0.5] \\ 1 + \sqrt{\varepsilon} & \text{if } x \in (0.5, 1] \end{cases}, \quad (30)$$

over the domain $\Omega = [0, 1]$, with periodic boundary conditions and $N = 1000$. The numerical solution is plotted in Fig. 5 for three sets of stiffness parameter and final time: $[\varepsilon, t_f] = \{[0.99, 0.1], [0.1, 0.05], [0.001, 0.007]\}$. The reference solution is computed with $N = 10000$ elements. Results show appropriate trends for the different Mach regimes.

5. Conclusion

In this paper, we propose a numerical scheme for the isentropic Euler system for all Mach regimes. The main element of the methodology is a penalization approach designed to handle the nonlinear stiff pressure term, which ultimately allows for a flux-splitting scheme that can be efficiently solved at the aid of an implicit-explicit time discretization. The numerical scheme is asymptotic preserving, and numerical results prove its accuracy and robustness for several test cases with different stiffness regimes. Future work will focus on the extension of the proposed methodology to free surface flows applications in the context of multi-phase models.

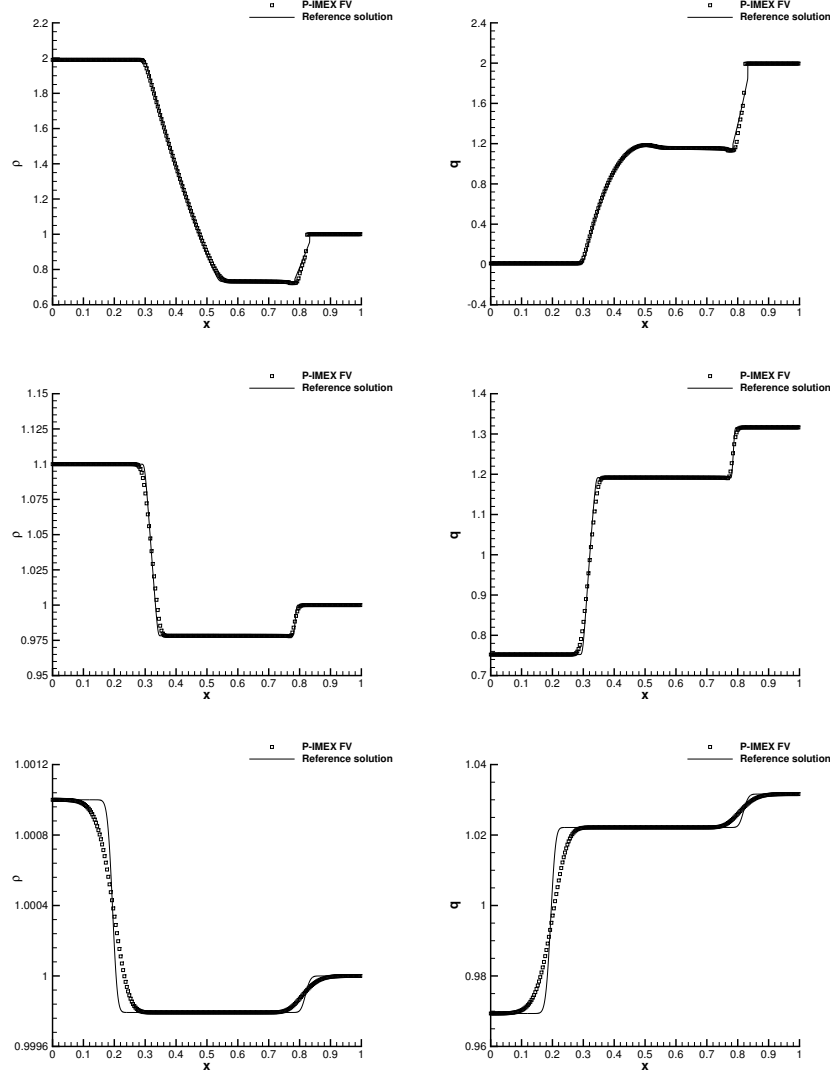


Fig. 5. Double rarefaction test problem at the final time with $N = 1000$ cells for $\varepsilon = 0.99$ (top row), $\varepsilon = 0.1$ (middle row) and $\varepsilon = 0.001$ (bottom row). Left column: density. Right column: momentum.

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